

# Nripesh

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## Education

### Vellore Institute of Technology (VIT)

B.Tech in Electrical and Electronics Engineering (CGPA: 8.46)

Vellore, Tamil Nadu

Sep 2022 – May 2026

**Relevant Coursework:** Power Systems Engineering, AC/DC Machines, Smart Grids, Control Systems, Industrial Automation, High Voltage Engineering, Power Electronics.

## Professional Experience

### NTPC Limited (Barh Super Thermal Power Station)

Vocational Trainee / Electrical Maintenance & Operations

Barh, Bihar

Jun 2025

- Analyzed operations of **660MW Supercritical Thermal units**, specifically Turbine-Generator synchronization and load dispatch.
- Monitored **Switchyard Operations (400kV/132kV)**, gaining practical exposure to Circuit Breakers, Isolators, and Transformers.
- Utilized **SCADA** systems to track real-time grid parameters, ensuring frequency stability.

### Maven Silicon

Embedded System Design Intern

Remote

Jun 2024 – Jul 2024

- Developed an IoT-based control system using **Embedded C** to simulate industrial safety cut-offs.
- Integrated Temperature and GPS sensors to process telemetry data with low latency, optimizing system response time.

## Key Projects

### Intelligent PI Control Based Smart Bidirectional EV Charging & V2G Coordination

Jan 2026 – May 2026

- Developed a comprehensive **MATLAB/Simulink** model for a price-responsive, bidirectional EV charging system featuring a 100 kVA inverter and an LCL filter while strictly maintaining 20–90% battery SOC limits.
- Engineered a centralized **Energy Management System (EMS)** enabling peer-to-peer energy transfer across a shared DC bus during high-price peak hours, completely bypassing expensive grid power.
- Evaluated an ultra-local model-free **Intelligent PI (iPI) controller**, achieving **60% faster settling times** and 3x lower overshoot under dynamic pricing conditions while keeping grid current THD below 3% (IEEE 519).

### Custom Water Leak Detector PCB

May 2026

- Engineered a custom hardware-based water leak detection circuit and printed circuit board from scratch using **KiCad 9.0**.
- Designed a reliable switching network utilizing an N-channel **2N7000 MOSFET** and a 10MΩ pull-down resistor to eliminate false triggering from ambient static electrical noise.

### Vehicle-to-Grid (V2G) Bidirectional Power Flow Simulation

Aug 2025 – Dec 2025

- Simulated a bidirectional power flow system where EVs act as mobile energy storage to stabilize grid frequency during peak loads.
- Designed control logic in **MATLAB/Simulink** allowing EV batteries to store excess energy and feed it back seamlessly.

### Smart Building Automation & Environment Control

Sep 2025

- Engineered an automated environment control system utilizing **ESP32** microcontrollers, PIR motion sensors, and LDR light sensors.
- Programmed logic in Embedded C++ for automatic load switching, reducing simulated energy consumption by 20%.

### Real-Time Face Recognition and Notification System Using AWS

Nov 2024

- Developed a serverless face recognition pipeline leveraging **Amazon S3, Lambda, and Rekognition** for secure facial metrics verification.
- Configured real-time notification alerts distributed instantly via **Amazon SNS** upon detecting known faces.

### Automotive Safety System

Jul 2024

- Designed an **Arduino-based** real-time tracking and emergency alert module utilizing GPS and GSM technologies.
- Engineered logic to instantly trigger and broadcast precise GPS coordinates to pre-registered contacts during critical fail-states.

## Technical Skills

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|------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Electrical Core:</b>            | Power System Analysis, AC/DC Machines, Switchgear & Protection, PCB Design, Power Electronics.                    |
| <b>Automation &amp; Control:</b>   | SCADA, PID / iPI Controllers, PLC Logic, IoT Sensor Integration, BMS Concepts.                                    |
| <b>Software &amp; Tools:</b>       | MATLAB, Simulink, Simscape, KiCad 9.0, AutoCAD (Electrical), LTSpice, AWS (S3, Lambda, Rekognition, SNS).         |
| <b>Hardware &amp; Programming:</b> | ESP32, Arduino, C/C++, Embedded C, Python.                                                                        |
| <b>Soft Skills:</b>                | Root Cause Analysis, Technical Troubleshooting, Industrial Safety, L <sup>A</sup> T <sub>E</sub> X Documentation. |

## Extracurricular & Achievements

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|------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <b>Team RotorFPV (Drone Development Club)</b>                                                                          | <b>VIT, Vellore</b>       |
| <i>Technical Member / Hardware Integration</i>                                                                         | <i>Jul 2024 – Present</i> |
| – Calibrated Flight Controllers and tuned <b>PID parameters</b> ; managed precision soldering of BLDC motors and ESCs. |                           |
| • <b>Winner</b> , ISDC Drone Challenge, BITS Goa (2025).                                                               |                           |
| • <b>1st Runner-Up</b> , Aero-Rush Technical Competition, NIT Trichy (2025).                                           |                           |
| • <b>2nd Runner-Up</b> , MultiRotor Competition, IIT Kanpur (2025).                                                    |                           |